# Cryptography tp 1 repport

## 1. Remote shell with SSH

in this part i used 2 computers running a linux system ( `arch` for the client and `openSUSE` for the server ), to access one's shell using the other with `ssh`.

### 1.a. Using password

- **command:**

```
$ ssh salim@192.168.1.19
```

- **output:**

```
(salim@192.168.1.19) Password:
Last login: Sun Feb 15 19:55:07 CET 2026 on tty1
Have a lot of fun...
salim@localhost:~
```

### 1.b. Using ssh key

- **command:**

```
$ ssh-keygen -t rsa
```

- **output:**

```
Generating public/private rsa key pair.
Enter file in which to save the key (/home/salim_belkacem/.ssh/id_rsa):
Enter passphrase for "/home/salim_belkacem/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/salim_belkacem/.ssh/id_rsa
Your public key has been saved in /home/salim_belkacem/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:IMIT5CKxtlriI9O+5B3wNFS5rW7CBScWmKfIciyhgoc salim_belkacem@archlinux
The key's randomart image is:
+---[RSA 3072]----+
|..oo  ..         |
|.=o.o..          |
|B*=oo..o         |
|Eo*++.o..        |
|+=o.o+ .S        |
|o+ + .o          |
|+oo.oo           |
|.=..o.o          |
|  +..o           |
+----[SHA256]-----+
```

- **command:**

```
$ ssh-copy-id salim@192.168.1.19
```

- **output:**

```
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/salim_belkacem/.ssh/
id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to
install the new keys
(salim@192.168.1.19) Password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'salim@192.168.1.19'"
and check to make sure that only the key(s) you wanted were added.``
```

- **command:**

```
$ ssh salim@192.168.1.19
```

- **output:**

```
Last login: Sun Feb 15 21:42:07 CET 2026 from 192.168.1.12 on ssh
Have a lot of fun...
salim@localhost:~>
```

## 2. file encryption with openssl

- **input**

```
$ echo Ceci est un test de chiffrement par AES via openSSL. > TestAES.odt
```

`TestAES.odt`:

```
Ceci est un test de chiffrement par AES via openSSL.
```

### 2.a. Encryption

- **input**

```
$ openssl enc -e -aes-256-cbc -in TestAES.odt -out TestAES_Crypted.odt
```

- **output**

```
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

*inserted password was: salim*

`TestAES_Crypted.odt`:

```
Salted__���Wo�w�W���x���DB�Z!�
                              ��3�f7��C$�<�B��n�5�n�\���0������
```

## 2.b. Decryption

- **input**

```
$ openssl enc -d -aes-256-cbc -in TestAES_Crypted.odt -out TestAES_Decrypted.odt
```

- **output**

```
enter AES-256-CBC decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
```

`TestAES_Decrypted.odt`:

```
Ceci est un test de chiffrement par AES via openSSL.
```